AgentWorkShop — System Design Philosophy & Functional Panorama (all elements code-anchored)

Design Principles

enforced in code, not slogans

1 · Single write throat
4-part write gate

2 · Fail-closed defense
uncertain → refuse · S1–S16

3 · Planning ≠ execution
LLM proposes · code acts

4 · Human authority
3 gates · never auto

5 · Evidence discipline
claims ≠ evidence

6 · Total auditability
decisions · events · journal

7 · Config-driven
drivers · templates · MCP

8 · Twin-first cutover
mirror → 6-step cutover

left = constitution
right = capabilities
truth: architecture-truth.md

Production-Line Operations

lines · templates · nodes · products · recipes · batches
start / stop with HITL approval
batch & recipe-window TOCTOU guards
channel line-binding (read-only escalation)

Acquisition & Control — DAQ / DCW

DAQ: modbus-tcp/rtu · opcua · mqtt · http → TSDB
vector / image frames → object store
DCW: graded writes · 3-tier ACK · journal anchors
S7 = explicit reject stub · serial = plugin

Agent Teamwork — Channels

LLM leads (omp / opencode / claude) + workers
108 governed host tools (all · selected)
execution modes: observe / manual / auto
budget · step limits · agent-token auth

Project Orchestration — SuperAgent

dual-graph plans (contract + execution)
route loop with no-LLM hot path
handoff bundles across channels
three-level acceptance L0 / L1 / L2

Optimization & Knowledge

DoE → AML training (ARX / static) → profiles
OSQP / SLSQP → governed dispatch
ontology 9 tables → 4 compiled artifacts
IDD diagnosis · RAG knowledge bridges

Observability & Analysis

dashboards · EDA / SPC / golden / OCAP
execution workbench (5 tabs)
HealthBoard · watchdog · error ledger
weekly reports · full data export

Deployment truths

single process · SQLite · TSDB · object store · MCP stdio 38 / HTTP 16 · LM Studio (local) or cloud